E-commerce Analytics

Context

This project analyzes transactional data from the Brazilian e-commerce platform Olist, covering orders between 2016 and 2018.

The objective is to evaluate commercial performance and understand customer behavior through data-driven analysis using SQL and Python.

Commercial Performance

The analysis of revenue and sales patterns revealed:

- Revenue is concentrated in a limited number of product categories
- Sales distribution varies significantly across regions
- A small subset of products and customers contributes disproportionately to total revenue

These patterns indicate a degree of concentration in the business, with certain segments driving a large portion of performance.

Customer Behavior

Customer behavior analysis shows a clear pattern of low engagement:

- The majority of customers make only a single purchase
- The average number of orders per customer is low
- Purchase frequency declines sharply as the number of orders increases

This indicates a predominantly transactional customer base, with limited recurring behavior.

Purchase Patterns

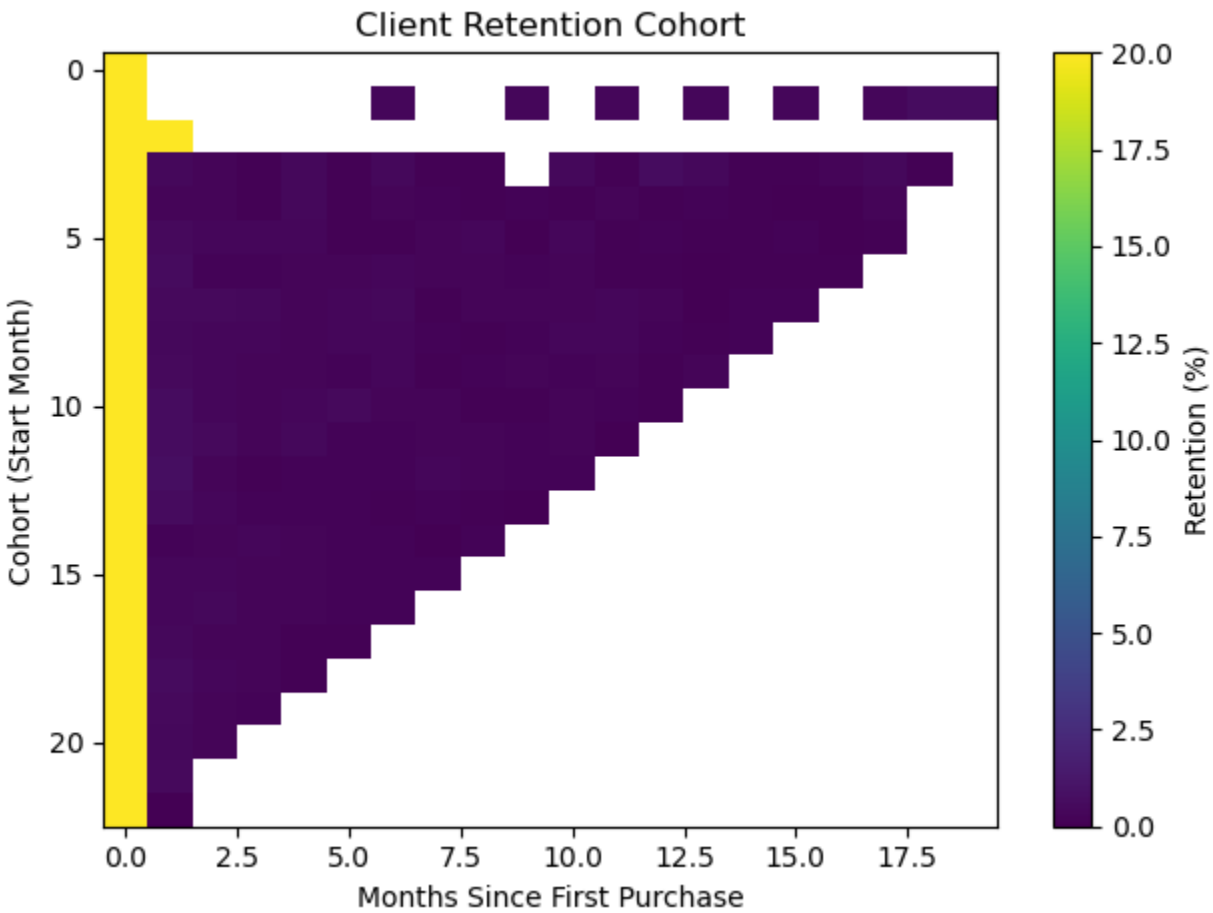
Temporal analysis of customer activity highlights:

- Long intervals between purchases among returning customers

- Significant variability in time between orders
- Low frequency of repeat purchases over time

These findings reinforce the absence of consistent purchasing cycles.

Retention Analysis (Cohort)



Cohort analysis reveals:

- A small level of repurchase immediately after the first order
- A sharp decline in retention in subsequent months
- Minimal long-term retention across customer cohorts

Although some short-term engagement exists, it is not sustained over time.

Key Takeaways

- Customer retention is the primary limitation in the current business model
- Most customers interact with the platform only once
- Recurring behavior is rare and not sustained
- Revenue depends heavily on initial purchases rather than long-term relationships

Business Implications

The results suggest opportunities to improve:

- Customer retention strategies
- Post-purchase engagement
- Loyalty and repeat purchase incentives
- Lifecycle marketing initiatives

Improving retention could significantly enhance long-term revenue stability.

Conclusion

The analysis provides a consistent view across commercial and behavioral metrics, indicating that while the platform generates revenue effectively, it struggles to retain customers over time.

Addressing retention and encouraging repeat purchases represents a key opportunity for business growth.

This report summarizes the key findings. For detailed analysis, including data processing, queries, and full visualizations, refer to the project notebooks.

Resumo (Português)

Este relatório apresenta a análise de dados do e-commerce Olist, com foco em desempenho comercial e comportamento dos clientes.

Os resultados indicam baixa recorrência na base, com predominância de compras únicas, intervalos elevados entre pedidos e baixa retenção ao longo do tempo.

Apesar de existir recompra no curto prazo, esse comportamento não se sustenta nos períodos subsequentes, indicando oportunidades de melhoria em estratégias de retenção e fidelização.

Este relatório apresenta um resumo dos principais resultados. Para uma análise detalhada, incluindo processamento de dados, consultas e visualizações completas, consulte os notebooks do projeto.